Offline

Dense Map

T_{gt}

T_{init}

π (K,[R|t])

D_{gt}

D_{init}

CNN F_{Dgt}

CNN F_{Dinit}

CNN F_{Ddist}

$I_{offline}$

RGB
(B, 3, H, W)

DINOv2 🔥F_I

Proj

CrossModalFusion

F_{fused}

HeatmapDecoder

H_{s1}

F

Sparse Map

GCM-Flow

E_{s1}

Σ

Pose Regression

q_0, t_0

GCM-Flow

E_{s2}

HeatmapDecoder

H_{s2}

Σ

Pose Regression

q_1, t_1

q_{gt}, t_{gt}

F Select Top-K salient points by heatmap score

Σ Element-wise multiply correspondence embedding with heatmap and sum over spatial dimensions

Online

Sparse Map

T_{init}

π (K,[R|t])

D_{dist}

CNN F_{Ddist}

I_{online}

RGB
(B, 3, H, W)

DINOv2 🔥F_I

Proj

HeatmapDecoder

H_{online}

GCM-Flow

E_{online}

Σ

Pose Regression

$\Delta\hat{T}$

q_0, t_0